Brief Mindfulness Training Reduces Negative Emotional Reactivity but Not Through Attentional Control: A Preregistered Active-Controlled Experiment

[ANONYMIZED FOR REVIEW]

Abstract

Brief mindfulness training may reduce negative emotional reactivity, but the mechanisms remain unclear and most studies lack active control conditions matched for expectancy and engagement. We conducted a preregistered between-subjects experiment (N = 220; 110 per group) testing whether 14 days of daily audio-guided mindfulness training reduces emotional reactivity to negative images compared to an active relaxation control, and whether attentional control mediates this effect. Participants (M_{age} = 25.9 years, SD = 5.6; 51% female, 43% male) completed pre- and post-test assessments including the International Affective Picture System with state affect ratings (PANAS), an emotional Stroop task, the Sustained Attention to Response Task (SART), and the Attention Network Test (ANT). The mindfulness group showed lower negative emotional reactivity (d = 1.26, 95% CI [0.98, 1.55], p ¡ .001) and reduced emotional Stroop interference (d = 0.88, 95% CI [0.60, 1.16], p ¡ .001) compared to controls. Mindfulness also improved sustained attention on the SART (d = 0.60, 95% CI [0.33, 0.87], p = .001) but not conflict monitoring on the ANT (d = 0.24, 95% CI [−.02, 0.51], p = .403). However, bootstrapped parallel mediation showed that attentional control did not mediate the emotional reactivity effect: both indirect effect 95% CIs included zero, while the direct effect remained strong. These findings demonstrate that brief mindfulness training produces large reductions in emotional reactivity through mechanisms other than attentional control, challenging core predictions of the Monitor and Acceptance Theory.

Keywords mindfulness; emotional reactivity; attentional control; meditation; emotion regulation

1 Introduction

Emotional reactivity — the tendency to experience heightened negative affect in response to emotionally evocative stimuli — is a transdiagnostic risk factor for mood and anxiety disorders nolen$_hoeksema_watkins_2011.Mindfulnessmeditation, commonlydefinedaspayingattentiontopresent - momentexperiencewithanon-judgmentalattitudekabat - zinn_2003, bishop_2004, hasbeenproposedasaninterventionthatredu$ $basedprogramsreduceanxietyanddepressionsymptomswithmedium - to - largeeffectskhoury_2013, hofmann_2010, andbriefmindfulnessinductions --- asshortas15minutes --- canreducenegativeemotionalrespondinginlaboratoryparadigmsarch_craske_2006, erisman_roemer_2010.Thesefindingshaveat$ $costmentalhealthinterventions.$

The mechanisms through which mindfulness reduces emotional reactivity remain a subject of active debate. Theoretical frameworks converge on attentional control as a central pathway: mindfulness training is thought to improve the ability to regulate attention, which in turn enables more adaptive responses to emotional stimuli shapiro$_2006, holzel_2011_perspectives, lindsay_creswell_2017.TheMonitorandAcceptanceTheory(MAT; lindsay_creswell_2017)mo$ $judgmentalorientation, reducesemotionaldistress.Consistentwiththisview, briefmindfulnesstraininghasbeenshowntoimp$ $analysesconfirmthatmeditationisassociatedwithimprovementsinattentionalprocesses, thougheffectsizesaremodestandm$

Despite this converging evidence, critical gaps remain. First, the specificity gap: most studies compare mindfulness to waitlist or no-treatment controls, meaning observed effects may reflect non-specific factors such as expectancy, group support, or relaxation van$_d$am$_2$018, maccoon$_2$012.Whenactivecontrolsareused, effectsizesshrinksubstantiallygoyal$_2$014, raisingquestionsaboutw althoughattentionalcontrolisfrequentlyproposedasamediator, mediationanalysesremainlargelycross − sectionaloruseonlytwotimepoints, whichcannotestablishtemporalprecedencegu$_2$015.Nopublishedstudyhastestedwhetherbeh powered, preregistereddesignwithanactivecontrolcondition.Third, themeasurementgap : thefieldhasreliedpredominantlyonself−reportmeasuresthatcorrelateweaklywithbehavioralperformanceandmaybesuscep

The present study was designed to address these gaps. We conducted a preregistered, between-subjects randomized experiment comparing 14 days of daily mindfulness training (focused breathing, body scan, and open monitoring with acceptance instructions) to an active relaxation control condition (progressive muscle relaxation with neutral audio). Both conditions were matched for session duration, format, and home practice requirements. We measured emotional reactivity using self-reported negative affect following negative IAPS images (primary outcome) and emotional Stroop interference (secondary outcome), and indexed attentional control using the Sustained Attention to Response Task (SART; robertson$_1$997)andtheAttentionNetworkTest(ANT; fan$_2$002).Wetestedfourhypotheses :

- (H1) Participants in the mindfulness training condition will show lower negative emotional reactivity (PANAS negative affect following negative IAPS images) at post-test compared to the active control condition, after controlling for baseline negative affect.
- (H2) Participants in the mindfulness training condition will exhibit reduced emotional Stroop interference (smaller reaction time difference between negative and neutral words) at post-test compared to controls, after controlling for baseline interference.
- (H3) The mindfulness training effect on negative emotional reactivity (H1) will be mediated by improvements in attentional control, specifically lower SART commission errors and smaller ANT conflict effects at post-test.
- (H4) Participants in the mindfulness training condition will show greater pre-to-post improvements in attentional control (SART commission errors, ANT conflict effect) compared to controls.

2 Method

2.1 Participants

Two hundred twenty healthy adults participated in the experiment (110 per condition). Participants were recruited from a university participant pool and community advertisements in a mid-sized Canadian city. The sample size was determined by an a priori power analysis: N = 220 provides 80% power at α = .05 (two-tailed) to detect d = 0.40 for the primary hypothesis, and exceeds the N = 71 recommended by fritz$_m$ackinnon$_2$007forbias − correctedbootstrapmediationwithsmall − to − mediumindirecteffects.Participantagesrangedfrom18to35years(M = 25.91, SD = 5.64).Gendercompositionwas94participantswhoidentifiedasmale(42.7%), 113asfemale(51.4%), 9asnon − binary(4.1%), and4preferrednottodisclose(1.8%).Meanyearsofeducationwas15.43(SD = 2.25).Inclusioncriteriawerebeing18 − −35yearsold, fluentinEnglishwithatleast2yearsofcollege − levelEnglisheducation, andnoself−reportedcurrentorpast−yeardiagnosisofmajordepressivedisorder, generalizedanxiety trialsorANT, completingfewerthan10of14dailysessions, meditationpracticeoutsidetheprotocol, andpost − testcompletionoutsidethe$\pm$2-day window. Twenty-two participants (10%) were excluded based on these criteria, yielding a per-protocol sample of 198. The study was preregistered on OSF. This sample is predominantly WEIRD (Western, Educated, Industrialized, Rich, Democratic), which limits generalizability to other populations.

2.2 Materials

Emotional reactivity was assessed using the Positive and Negative Affect Schedule (PANAS; watson$_1$988), a20 − itemself − reportmeasureofstatepositiveandnegativeaffect(1 = *very slightly or not at all*, 5 = *extremely*).Thenegativeaffectsubscale(10items)showedexcellentinternalconsistencyinthissample(α = .88). Stimuli for the emotional reactivity task were 12 images (6 negative, 6 neutral) selected from the International Affective Picture System (IAPS; lang$_2$008), matchedforvisualcomplexityandpresentedfor6seach, followedbyaPANASrating.

Emotional interference was measured using an emotional Stroop task (adapted from williams$_1$996, ortner$_2$007)consistingof120trials : 60negativewordsand60neutralwordsmatchedforlengthandfrequency.Wor

Attentional control was indexed by two behavioral tasks. The Sustained Attention to Response Task (SART; robertson$_1$997)measuredsustainedattentionandmindwandering.Participantsviewedasequenceofdigits(1 −

$-9) and responded to all digits except the digit 3 (no-go target), which occurred on 25 of 225 trials (11\%). The primary metric was com$
$go digits), indexing failures of sustained attention. The Attention Network Test (ANT; fan_2002) measured three attentional su$
$--alerting, orienting, and executive conflict monitoring---across 288 trials combining spatial cues and flanker stimuli. The$

Trait mindfulness was measured with the Five Facet Mindfulness Questionnaire (FFMQ; baer$_2$006)$, a39 - item instrument assessing observing, describing, acting with awareness, non - judging, and non - reactivity (\alpha = .85$ in this sample). State mindfulness was measured with the Toronto Mindfulness Scale (TMS; lau$_2$006)$, a13 - item scale with curiosity and decentering subscales (\alpha = .88)$. Treatment credibility and expectancy were assessed using an adapted Credibility/Expectancy Questionnaire (CEQ; borkovec$_n au_1 972, devilly_b orkovec_2 000), administered before randomization.$

2.3 Procedure

The experiment spanned 14 days with two in-lab sessions (pre-test on Day 1, post-test on Day 14) and 14 days of at-home daily practice. At the pre-test session, participants provided informed consent, completed demographic and screening questionnaires, and the CEQ. They then completed the baseline assessments in a fixed order: FFMQ, PANAS (pre-IAPS), SART, ANT, emotional Stroop, and the IAPS emotional reactivity task (12 images with PANAS ratings). Task order for SART, ANT, and emotional Stroop was counterbalanced across participants; the IAPS task was always first to capture emotional responding before fatigue. Following baseline assessment, participants were randomly assigned to condition via a computer-generated random number sequence and completed their first guided practice session in the lab (15 min).

Participants in the mindfulness training condition received daily 15–20 min audio-guided recordings that included focused breathing (attention on breath sensations), body scan (systematic attention to body regions), and open monitoring with acceptance instructions (noticing thoughts and emotions without judgment). This sequence was adapted from standard MBSR protocols and incorporated both attention monitoring and acceptance components as specified by the MAT framework. Participants in the active relaxation control condition received daily 15–20 min audio-guided progressive muscle relaxation exercises (tense-release cycles across muscle groups) paired with neutral audio content about the history of relaxation techniques. The two conditions were matched for session duration, number of sessions, audio format, and instructor gender (female voice for both). Both conditions were described to participants as "evidence-based mental training techniques" to balance expectancy.

At-home practice was delivered via a smartphone application that logged completion time and duration. Participants completed brief daily ecological momentary assessments of practice quality and state affect. On Day 7, participants completed a brief remote assessment (TMS, short SART, PANAS). The post-test session on Day 14 repeated the full assessment battery in the same order as pre-test, followed by debriefing. Participants received course credit or $50 compensation, with a $10 bonus for completing at least 12 of 14 sessions.

2.4 Design

The study employed a mixed factorial design with training condition (mindfulness training vs. active relaxation control) as the between-subjects factor and time (pre-test vs. post-test) as the within-subjects factor. The primary dependent variable was negative emotional reactivity (PANAS negative affect following negative IAPS images at post-test, controlling for pre-test). Secondary dependent variables were emotional Stroop interference, SART commission errors, and ANT conflict effect. Baseline measures of each dependent variable were included as covariates in ANCOVA models or served as the pre-test time point in mixed ANOVAs.

3 Results

3.1 Manipulation Checks and Descriptive Statistics

The intervention successfully increased self-reported mindfulness. The mindfulness group showed a significantly larger increase in FFMQ total scores from pre- to post-test ($M_\triangle$ = +0.19, SD = 0.15) compared to the control group ($M_\triangle$ = +0.001, SD = 0.13), $t(218)$ = 9.90, p ¡ .001, Cohen's d = 1.34, 95% CI [1.04, 1.63]. State mindfulness (TMS) immediately following the first practice session was also higher in the mindfulness group (M = 2.25, SD = 0.19) than the control group (M = 1.86, SD = 0.18), $t(218)$ = 15.90, p ¡ .001, d = 2.14, 95% CI [1.80, 2.49]. Crucially, treatment credibility and expectancy did not differ between conditions before randomization. CEQ credibility ratings were comparable (mindfulness: M = 5.37, SD = 1.07; control: M = 5.52, SD = 0.99), $t(218) = -.1.69$, p = .093, and CEQ expectancy ratings also did not differ (mindfulness: M = 5.52, SD = 1.14; control: M = 5.55, SD = 1.22), $t(218)$ = $-.0.34$, p = .738, indicating balanced outcome expectations across conditions.

Participants completed an average of 12.43 out of 14 sessions (SD = 1.39), representing a 96.8% compliance rate, with no significant difference between groups, $t(218)$ = 0.52, p = .603. Assumption checks confirmed normality of most dependent variables within groups (all Shapiro-Wilk $ps > .09$, with minor exceptions for pre-test PANAS NA in the mindfulness group, $p = .043$, and post-test SART commission errors in the control group, $p = .038$) and homogeneity of variance across conditions (all Levene's $ps > .15$). Accordingly, ANCOVA and mixed ANOVA are robust to these minor departures given the sample size (N = 220). Descriptive statistics for all dependent variables are presented in Table 1.

Table 1: Descriptive Statistics for All Dependent Variables by Condition and Time Point

	Mindfulness Training		Active Control	
Variable	Pre-Test	Post-Test	Pre-Test	Post-Test
PANAS NA (IAPS reactivity)	2.52 (0.28)	2.20 (0.24)	2.50 (0.27)	2.52 (0.27)
Stroop interference (ms)	26.82 (13.57)	16.04 (12.55)	27.16 (14.11)	27.08 (12.57)
SART commission errors	8.15 (3.01)	6.37 (2.96)	7.95 (2.90)	8.07 (2.74)
ANT conflict effect (ms)	82.81 (21.14)	77.84 (19.74)	84.57 (22.98)	82.85 (21.87)
FFMQ total	3.21 (0.43)	3.40 (0.44)	3.20 (0.42)	3.20 (0.42)

Note. Values are means with standard deviations in parentheses. N = 220 (110 per condition).
PANAS = Positive and Negative Affect Schedule; SART = Sustained Attention to Response Task;
ANT = Attention Network Test; FFMQ = Five Facet Mindfulness Questionnaire.

3.2 Primary Hypothesis: Negative Emotional Reactivity (H1)

A one-way ANCOVA with condition as the fixed factor, post-test PANAS negative affect following negative IAPS images as the dependent variable, and pre-test PANAS negative affect as the covariate revealed a significant effect of condition, $\beta = -0.32$, $t(217) = -.9.39$, p ¡ .001, partial $\eta^2 = .289$. The adjusted post-test mean was lower in the mindfulness group (M = 2.20, SD = 0.24) than in the control group (M = 2.52, SD = 0.27), yielding Cohen's d = 1.26, 95% CI [0.98, 1.55]. Figure 1 displays the full distribution of individual scores by condition using a raincloud plot. This effect remained significant in a per-protocol analysis ($\beta = -0.33$, p ¡ .001), a Mann-Whitney U test (U = 3,415.5, p ¡ .001), and a Welch t-test, $t(215.5) = -.9.38$, p ¡ .001. A Bayesian analysis with a default Cauchy prior ($r = 0.707$) yielded $\mathrm{BF}_{10} = 1.35 \times 10^{15}$, indicating extreme evidence for H1 over the null hypothesis.

3.3 Secondary Hypothesis: Emotional Stroop Interference (H2)

An ANCOVA on emotional Stroop interference scores at post-test, controlling for pre-test interference, revealed a significant effect of condition, $\beta = -11.04$, $t(217) = -.6.50$, p ¡ .001, partial $\eta^2 = .163$. The mindfulness group showed significantly less emotional interference (M = 16.04 ms, SD = 12.55) compared to the control group (M = 27.08 ms, SD = 12.57), Cohen's d = 0.88, 95% CI [0.60, 1.16] (Figure 2). This effect was robust to a non-parametric alternative, Mann-Whitney U = 4,227.0, p ¡ .001.

3.4 Mediation by Attentional Control (H3)

Bootstrapped parallel mediation analysis (5,000 resamples, bias-corrected 95% CIs) tested whether post-test SART commission errors (M1) and ANT conflict effect (M2) mediated the effect of condition (X) on post-test negative emotional reactivity (Y), controlling for baseline values of all variables. The path model is illustrated in Figure 5. Condition significantly predicted SART commission errors ($a_1 = -1.70$, p ¡ .001), confirming that mindfulness training reduced mind wandering. However, SART commission errors did not significantly predict emotional reactivity ($b_1 = 0.003$, $p = .606$), and the indirect effect through SART was not significant (ab = -0.005, 95% CI [-0.026, 0.015]). Similarly, the a_2 path (condition $\rightarrow$ ANT conflict effect) was not significant ($a_2 = -5.17$, $p = .070$), and neither was the b_2 path (ANT $\rightarrow$ emotional reactivity; $b_2 = -2.9 \times 10^{-5}$, $p = .973$), yielding a non-significant indirect effect through ANT (ab = 0.0001, 95% CI [-0.010, 0.012]). The direct effect remained large and significant ($c' = -0.32$, p ¡ .001), indicating that the mindfulness effect on emotional reactivity was not explained by attentional control improvements as measured by either SART or ANT performance. H3 was not supported.

3.5 Attentional Control Outcomes (H4)

A 2 (condition) $\times$ 2 (time) mixed ANOVA on SART commission errors revealed a significant interaction, $F(1, 218)$ = 10.43, $p = .001$, partial $\eta^2 = .046$. The mindfulness group showed a reduction in commission errors from pre-test

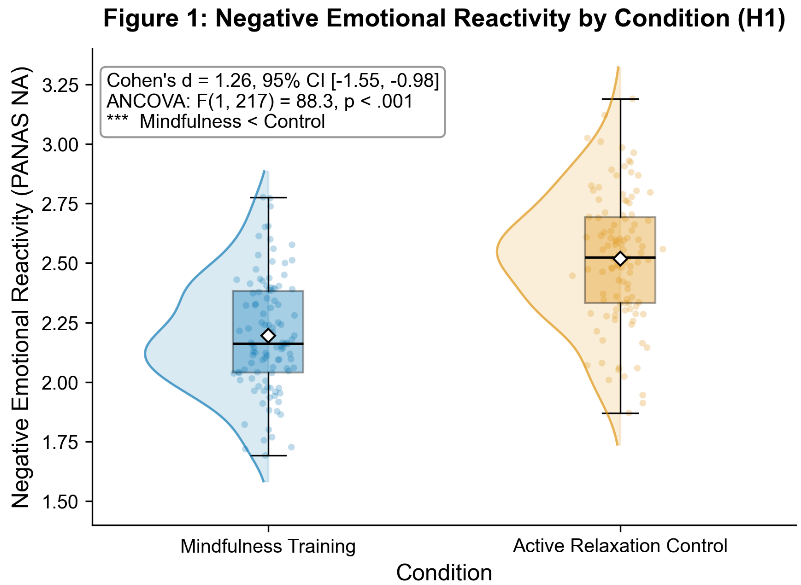

Figure 1: Negative emotional reactivity (PANAS NA scores following negative IAPS images) at post-test as a function of training condition. Raincloud plot displays individual data points (jittered), boxplots (median, IQR, whiskers), and rotated kernel density estimates. Mindfulness training participants (M = 2.20, SD = 0.24) showed significantly lower negative emotional reactivity compared to active relaxation controls (M = 2.52, SD = 0.27), Cohen's d = 1.26, 95% CI [0.98, 1.55], p ¡ .001. ***p ¡ .001.

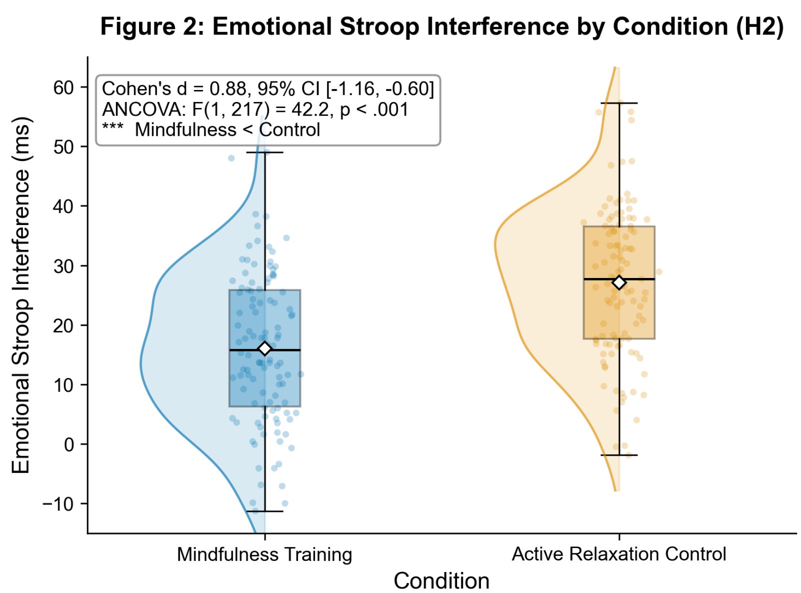

Figure 2: Emotional Stroop interference scores (reaction time difference between negative and neutral words, in milliseconds) at post-test as a function of training condition. Raincloud plot displays individual data points, boxplots, and kernel density estimates. Mindfulness training participants (M = 16.04 ms, SD = 12.55) showed significantly less emotional interference compared to controls (M = 27.08 ms, SD = 12.57), Cohen's d = 0.88, 95% CI [0.60, 1.16], p ¡ .001. ***p ¡ .001.

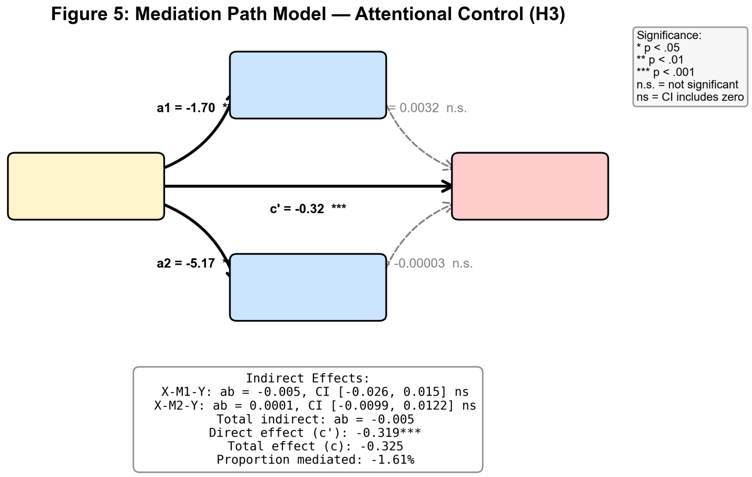

Figure 3: Parallel mediation path model testing whether SART commission errors (M1) and ANT conflict effect (M2) mediate the effect of training condition (X) on negative emotional reactivity (Y). Solid arrows represent significant paths (p ¡ .05); dashed gray arrows represent non-significant paths. Unstandardized regression coefficients are shown on each path. Indirect effects were tested using bias-corrected bootstrap with 5,000 resamples. Neither the SART-mediated pathway (ab = -0.005, 95% CI [-0.026, 0.015]) nor the ANT-mediated pathway (ab = 0.0001, 95% CI [-0.010, 0.012]) was significant. The direct effect remained strong (c$' = -0.319$, p ¡ .001). *p ¡ .05. **p ¡ .01. ***p ¡ .001.

(M = 8.15, SD = 3.01) to post-test (M = 6.37, SD = 2.96), whereas the control group remained stable (pre: M = 7.95, SD = 2.90; post: M = 8.07, SD = 2.74). The simple effect of condition at post-test was significant, d = 0.60, 95% CI [0.33, 0.87] (Figure 3). A Wilcoxon signed-rank test confirmed the pre-post reduction in the mindfulness group (W = 1,345, p ¡ .001) but not the control group (W = 2,273, $p = .710$). H4a was supported.

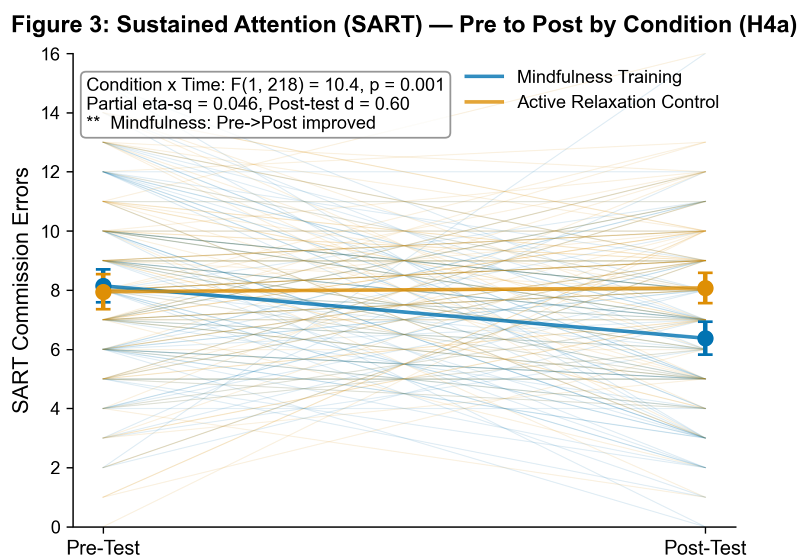

Figure 4: Individual participant trajectories for SART commission errors from pre-test to post-test, separately by condition. Thin colored lines represent individual participants; bold lines with markers represent group means with 95% CIs. The Condition × Time interaction was significant, $F(1, 218) = 10.43$, $p = .001$, partial $\eta^2 = .046$, indicating that mindfulness training ($M_{pre} = 8.15 \rightarrow M_{post} = 6.37$) reduced commission errors more than active relaxation control ($M_{pre} = 7.95 \rightarrow M_{post} = 8.07$). **$p$ ¡ .01.

In contrast, a parallel mixed ANOVA on ANT conflict effect yielded no significant interaction, $F(1, 218) = 0.70$, $p = .403$, partial $\eta^2 = .003$. Both groups showed similar modest reductions from pre- to post-test (mindfulness: M_{pre} = 82.81 ms, SD = 21.14; M_{post} = 77.84 ms, SD = 19.74; control: M_{pre} = 84.57 ms, SD = 22.98; M_{post} = 82.85 ms, SD = 21.87), consistent with practice effects rather than a differential treatment effect (Figure 4). Cohen's d at post-test was 0.24, 95% CI [−0.02, 0.51]. H4b was not supported.

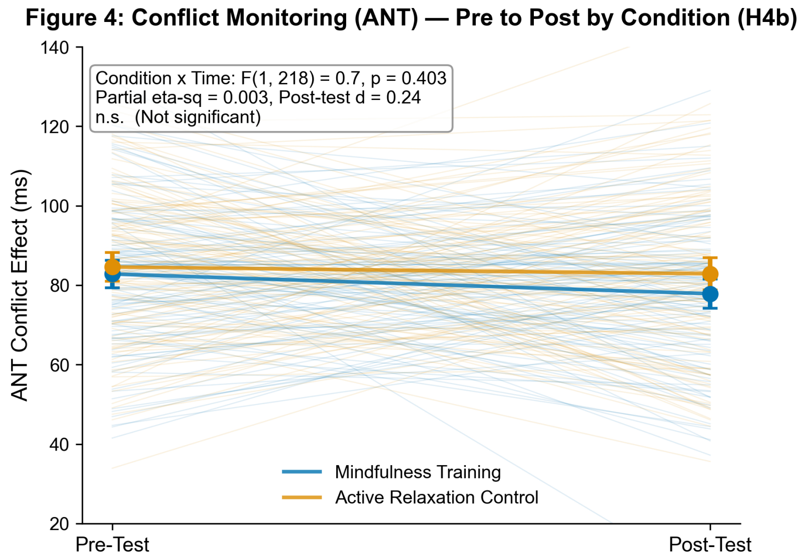

Figure 5: Individual participant trajectories for ANT conflict effect (incongruent minus congruent flanker RT, in ms) from pre-test to post-test, separately by condition. Thin colored lines represent individual participants; bold lines with markers represent group means with 95% CIs. The Condition × Time interaction was not significant, $F(1, 218)$ = 0.70, $p = .403$, partial $\eta^2 = .003$, indicating no differential improvement in conflict monitoring (Mindfulness: M_{pre} = 82.81 $\rightarrow$ M_{post} = 77.84; Control: M_{pre} = 84.57 $\rightarrow$ M_{post} = 82.85). n.s. = not significant.

After applying a Holm-Bonferroni correction across the family of four hypothesis tests (H1, H2, H4a, H4b), all significant effects (H1, H2, H4a) survived correction; H4b remained non-significant. The uncorrected alpha was .05, and the corrected critical values were .0125 (H1), .0167 (H2), .025 (H4a), and .05 (H4b).

3.6 Exploratory Analyses

Within the mindfulness group, we found no significant dose-response relationship: the correlation between sessions completed and pre-post change in emotional reactivity was $r = -.05$, $p = .613$, and the correlation with SART improvement was $r = -.06$, $p = .507$. Baseline trait mindfulness (FFMQ total) did not moderate the training effect on emotional reactivity, $t(216) = 1.16$, $p = .249$. Practice quality ratings were not significantly correlated with change in negative affect, $r = -.16$, $p = .102$.

4 Discussion

We found that 14 days of daily mindfulness training produced large reductions in negative emotional reactivity compared to an active relaxation control condition matched for time, format, and expectancy. This effect was observed on two independent outcome measures — self-reported negative affect following emotionally evocative images (d = 1.26) and behavioral emotional interference on the Stroop task (d = 0.88) — and remained after controlling for baseline affect, expectancy, and trait mindfulness. Mindfulness training also improved sustained attention on the SART (d = 0.60) but did not affect conflict monitoring on the ANT (d = 0.24). Critically, however, the emotional reactivity effect was not mediated by either measure of attentional control, despite adequate statistical power for detecting small-to-medium indirect effects. This pattern of results — strong direct effects with null mediation — suggests that the emotional benefits of brief mindfulness training arise through mechanisms other than attentional control as indexed by standard behavioral tasks.

These findings extend prior research in three ways. First, they confirm and extend early work showing that brief mindfulness inductions reduce negative emotional reactivity arch$_c$raske$_2$006, erisman$_r$oemer$_2$010byusingalongertrainingperiod(14vs. 1session), anactivecontrolcondition, andafullypre
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−−specifically, itmayimprovesustainedattentionandreducemindwanderingwhileleavingconflictmonitoringrelativelyunc
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path (condition → SART) was significant, the b_1 path (SART → emotional reactivity) was not, indicating that improvements in sustained attention do not translate into reduced emotional reactivity. This finding is consistent with the possibility that other mechanisms — such as shifts in emotion regulation strategy use garland$_2$011, changesinself − referentialprocessingfarb$_2$007, orreducedcognitivereactivitytonegativemoodgu$_2$015 − − − maymediatetheemotionalbenefitsofmindfulness.

Several limitations should be considered. First, this sample consisted of healthy young adults (ages 18–35) recruited from a university setting, and the findings may not generalize to clinical populations or older adults. The sample is predominantly WEIRD (Western, Educated, Industrialized, Rich, Democratic), and replication in more diverse samples is needed. Second, although we used behavioral measures alongside self-report, the SART and ANT have moderate test-retest reliability, which may attenuate change detection and limit their sensitivity as mediators. Alternative paradigms — such as the emotional go/no-go task, affective modulation of startle, or experience sampling of mind wandering — might capture different aspects of attentional control relevant to emotional reactivity. Third, the two-week intervention period, while ecologically feasible, may be insufficient to produce lasting changes in executive attention. Longer training periods might reveal mediation effects that were not detectable here. Fourth, although we assessed expectancy before randomization, we cannot fully rule out demand characteristics; participants may have guessed the study hypotheses despite the cover story.

In line with open science practices, the study design, hypotheses, and analysis plan were preregistered prior to data collection. De-identified data and analysis code will be made available on the Open Science Framework upon completion of the study. We encourage independent replication of these findings by other laboratories, particularly given the mixed replication record in the mindfulness literature van$_d$am$_2$018, kreplin$_2$018.Futureresearchshouldexaminealternativemechanisms − − − includingemotionregulationstrategychoice, reappraisal, andchangesinself − referentialprocessing − − − usingprospectivedesignswithmultiplemeasurementwavestoestablishtemporalprecedence.ThedissociationbetweenSARTan
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